

Portfolio optimization in electricity markets

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Abstract

In a competitive electricity market, Generation companies (Gencos) face price risk and delivery risk that affect their profitability. Risk management is an important and essential part in the Genco's decision making. In this paper, risk management through diversification is considered. The problem of energy allocation between spot markets and bilateral contracts is formulated as a general portfolio optimization problem with a risk-free asset and n risky assets. Historical data of the PJM electricity market are used to demonstrate the approach.

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1. Introduction

Deregulation in the electricity industry has introduced competitive markets. Generation companies (Gencos) no longer enjoy guaranteed rate of return as in the old regulated environment. The price of electricity Gencos receive in a competitive market depends on many factors: bidding prices of all market participants, load demand, unit outages, etc. It is uncertain and volatile. There is usually more than one market for a Genco to enter. Gencos are faced with the prospect of making more profit or the risk of losing money. The scheduling decisions of Gencos are important in determining their profitability. Recognizing market risk and management of such risks are essential for Gencos in a competitive market.

Risk refers to the possibility of suffering harm or loss; danger or hazard. Risks result from uncertainty. However, there is a difference between risk and uncertainty: risk is something that usually can be controlled whereas uncertainty is beyond anybody's control. In the electricity market, the profits of Gencos are influenced by many uncertain factors: unit outage, other genco's bidding strategy, congestion in transmission, demand change, etc. These uncertainties bring about risks in electricity pricing and delivery. Risks of spot price volatility in electricity markets are especially significant. Operating data have shown that daily

spot price volatility in electricity is much higher than that of any other commodity. The main reason for this may be attributed to the particular characteristic of non-storability of electricity.

Risk management is the process of achieving a desired return/profit, taking into considerations of risks, through a particular strategy. In the financial field, there are two means to control risk. One is through risk financing by using *hedging* to offset losses that can occur and the other is through risk reduction using diversification to reduce exposure to risks. Instruments for risk management include forward contracts, futures contracts, options, etc. Forward contracts are agreements to buy/sell an agreed amount of the commodity at a specified price at a designated time. Futures contracts are standardized forward contracts that are traded on exchange and no physical delivery is necessary. Options are contracts that provide the holder the right but not the obligation to buy/sell the commodity at a designated time at a specified price. Hedging is to use these financial instruments with the payoff patterns to offset the market risks. Diversification is to engage in a wide variety of markets so that the exposure to the risk of any particular market is limited. Applying this concept to energy trading in an electricity market, diversification means to trade energy through different physical trading approaches.¹ In the energy market, both physical trading approach (e.g., spot

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¹ Physical trading approach refers to the trading approach in which actual physical energy are traded while financial trading approach only involves financial settlement, no actual physical energy are traded through financial trading.

market, contract market) and financial trading approach (e.g., futures contracts, options, swaps, etc.) are available. A combination of these trading approaches is defined as a *portfolio* and the corresponding risk-control methodology is called *portfolio optimization*.

A commonly adopted measure for risk assessment, i.e., assessing risk exposure of financial portfolios, is the Value at Risk (VaR), which is the monetary value that the portfolio will lose less than that amount over a specified period of time with a specified probability.

Various aspects of risk management have been applied to the electricity market. Different forward contracts that can provide hedging to the risk of spot prices for market participants are proposed [1–4]. The usefulness of the application of futures contracts in an electricity market is demonstrated in [5–8]. Valuation of different contracts is considered in [9–11]. Monte Carlo simulation and decision analysis have been applied to find the optimal contract combination [12–15]. Various issues related to the combined spot/bilateral-contract dispatch are investigated in [16–18]. VaR has been applied to risk assessment in electricity markets [19–22]. Concepts from financial option theory have been utilized in the valuation of generation assets [23–25].

We are addressing the problem of trading scheduling for a Genco, i.e., to optimally use both physical trading approaches and financial trading approaches to maximize its profit potential, taking into consideration the associated risk factors. It involves the optimal allocation of the Genco's output energy among multiple markets (e.g., spot market, contract market, futures market, etc.) with the objective of maximizing its benefit and minimizing the corresponding risk. We apply the approaches of portfolio optimization in Modern Portfolio Theory (MPT) [26] to the problem. The method explicitly considers decision-makers' risk aversion and the statistical correlation among alternative outcomes. Although MPT is widely known in the financial literature, its application in electricity markets might be of interest. The reason is that electricity contracts have different risk characteristics under different electricity markets with different pricing systems, which is due to the congestion in transmission. It is further explained in Section 2 through the introduction of trading environment in electricity markets. Only price risk, delivery risk and physical trading approaches are considered in this paper. Price risk due to spot market fluctuations and delivery risk due to transmission congestion are related to power system operation.

In terms of applications, an electricity spot market that adopts uniform marginal pricing scheme displays only price risk and a spot market that adopts locational marginal pricing or zonal pricing displays not only price risk but also delivery risk. In the following, Section 2 introduces the background of the electricity market with different pricing system which includes the trading environment and associated risks. Section 3 describes the basic theory and methodology to portfolio optimization which can be applied to electricity markets with different pricing system. Section 4 develops an approach to energy allocation among physical trading approaches, i.e., spot market and contract market. Example demonstrates the proposed energy allocation method using historical data of the PJM market. Finally, Section 5 concludes the paper.

2. Electricity markets

Most electricity markets provide two types of markets in which energy is traded: the spot market and the (physical) forward market [27]. In the contract market, Gencos trade energy by way of signing contracts, which are referred to as physical forward contracts, with their counterparters (e.g., energy consumers). Specific details such as trading quantity (MW), trading duration (h), trading price (\$/MWh) and delivery point are bilaterally negotiated between Gencos and consumers or their agents. Bilateral contracts are signed before the actual trading period. In other words, trading quantity and price are set in advance. Physical forwards can be traded on an exchange or in a bilateral manner through over the counter (OTC)² transactions.

As for the spot market, in this paper, we adopt FERC definition in its standard market design [28] that all the energy traded in the real-time and day-ahead market as spot energy. The common ground among these markets is that they all involve a centralized auction mechanism, by ISO, RTO or any such organization, to determine which generation units should be deployed and how much energy each selected unit should produce to meet the demand. From a Genco's point of view, selling energy in the spot market means to submit a bid (price and quantity) to the exchange (Power Pool/ISO) and get either of the two alternative results: (1) the exchange accepts the Genco's bid and pays the Genco the market clearing price (MCP) for its actual energy output; or (2) the exchange rejects the Genco's bid, i.e., the Genco sells nothing in the spot market. The MCP depends on everybody's bids, as well as the load demand, and is therefore uncertain. Three types of pricing systems have been adopted in the spot market: uniform marginal pricing, zonal pricing and nodal pricing (or locational marginal pricing (LMP)).

Uniform marginal pricing was adopted in the earlier England and Wales market and is followed by many markets around the World. In such a market, only one energy price is used for ex post settlement for each trading interval. Uniform marginal pricing makes it easier to achieve market liquidity. A Genco, on the other hand, can make certain of its revenue by signing bilateral contracts with its customers at fixed energy prices. Hence, bilateral contracts can be considered as risk-free transactions if the Genco's production cost is assumed deterministic.³ Trading schedule in an electricity market with uniform pricing is to optimally allocate energy between risky spot market and risk-free contract market.

In a zonal pricing system, an interconnected power system is partitioned into pre-defined geographical areas, called zones, based on the knowledge that limitations in transmission exist between these zones.⁴ When there is no congestion, one uni-

² OTC is a kind of derivatives market in which non-standard products (e.g., contracts) are traded. Trades on the OTC market are negotiated directly through dealers.

³ Only fuel-fired plants are considered and fuel prices are assumed deterministic in order to focus on the risk of electricity prices in this paper.

⁴ Transmission system has operating constraints that limit the maximum amount of power that are allowed to flow through transmission lines. The limit is set either by conductor thermal loading limit or by system stability consid-

form MCP is used throughout the system. When congestion occurs, the power system is split into two zones with congested lines in between and each zone has its own MCP, called zonal prices. Transmission congestion limits the amount of generation on the cheaper MCP to supply the demand that pays more on the more expensive MCP side. The product of the price difference between the zones involved and the trading amount is called the congestion charge, which has to be added to the trading cost for the bilateral trading. Congestion and the resulting zonal prices are uncertain and unpredictable, which makes inter-zonal bilateral contracts risky. Only intra-zonal contracts in such a market are risk-free. Zonal pricing was used in pre-2002 California market and is still used in Nordic and some other markets.

Locational marginal (or nodal) price, as the name implies, is the marginal price at each location or node of a power system. LMPs are obtained when nodal power balance equations (power flow equations), as well as transmission line loading limits, are explicitly incorporated in the welfare maximization problem (optimal power flow or OPF). When there is no congestion, there is one price in the interconnected system if transmission losses are ignored. When any one of the transmission line is congested, the marginal energy prices will vary from locations. LMP is recommended by FERC in its standard market design [28] and adopted by many RTOs and ISOs [29–32]. Same as in zonal pricing, the difference between locational prices represents congestion charges to the bilateral trading. Since demand, generation pattern and transmission loading vary over time, network congestion, locational prices, and congestion charges also vary over time and are uncertain. Except contract with customers at the same location, all contracts are risky. Note that from theoretical point of view, methodology for energy trading scheduling under LMP and zonal pricing is the same. Our presentation will be stated in terms of LMP, knowing that it is equally applicable to zonal pricing. Besides, trading schedule in a uniform pricing market is just a special example in the LMP market. A market with LMP is therefore supposed general in this paper.

3. Portfolio optimization

3.1. Modern Portfolio Theory

The Modern Portfolio Theory [26] is principles underlying analysis and evaluation of rational portfolio choices based on risk-return trade-offs and efficient diversification. In other words, MPT is an approach to measuring the risk of an asset, quantifying trade-off between risk and expected return, and finally forming an optimal portfolio of assets. Markowitz is the father of MPT. His original article [33] and book [34] on the subject clearly depicted, for the first time, Modern Portfolio Theory. The book was filled with insights and suggestions that anticipated many of the subsequent developments in the

field. The important message of the theory was that asset could not be selected only on characteristics that were unique to the security. Rather, an investor had to consider how each security co-moved with all other securities. Furthermore, taking these co-movements into account resulted in an ability to construct a portfolio that had the same expected return and less risk than a portfolio constructed by ignoring the interactions between securities. Portfolio theory is a well-developed paradigm. There are excellent textbooks such as [26] and [35] on this subject. There are also good reviews in more advanced texts such as [36]. Finally, there are good review articles such as [37].

In the following, we formulate the general portfolio optimization problem as a quadratic programming problem. We also give a brief tutorial treatment of the standard approach in financial theory.

3.2. Portfolios of assets

A portfolio of assets is a combination of all potential assets. Let indexes $1-n$ denote risky assets, while index $n+1$ denotes the risk-free asset. All risk-free assets can be accumulated into one risk-free asset. Given each asset's rate of return (return for short), r_i ($i=1-n+1$), the portfolio's return (denoted by r_C) is the weighted average of the component asset return with the investment proportions as weights (denoted by w_i), i.e., $r_C = \sum_{i=1}^{n+1} w_i r_i$. The portfolio consists of risky assets and is also risky. In other words, its return (r_C) is uncertain. Let us assume that we do have some knowledge of the situation in terms of the probability distribution of the outcomes. The mean of the probability distribution of the return, or the expected return, is an indication of the expected profitability. The variance of the distribution indicates how wide spread is the possible outcomes around the mean. The larger is the variance, the more uncertain is the outcome. Therefore, the variance or the standard deviation of the distribution can be used as an indication of the risk involved. This is the basis of the mean-variance criterion (MVC) used by Markowitz and Tobin to develop the modern theory of investment choice under uncertainty.

The expected return ($E(r_C)$) and its variance ($\sigma^2(r_C)$) can be expressed as follows:

$$E(r_C) = \sum_{i=1}^{n+1} w_i E(r_i) \quad (1)$$

$$\sigma^2(r_C) = \sum_{i=1}^{n+1} \sum_{j=1}^{n+1} w_i w_j \sigma_{ij} = \sum_{i=1}^{n+1} w_i^2 \sigma_i^2 + \sum_{j \neq i} \sum_{i=1}^{n+1} w_i w_j \sigma_{ij} \quad (2)$$

where $\sum_{i=1}^{n+1} w_i = 1$, $w_i \geq 0$, σ_i^2 is the variance of the return on the i th asset; σ_{ij} is the covariance between the returns on the i th asset and the j th asset. The covariance σ_{ij} measures how many the returns on two assets move in tandem.

3.3. Portfolio selection

According to the MVC, a decision that results in higher expected return and lower risk would be preferred. In other

erations. When transmission constraints limit power generation from otherwise economic considerations, we say the transmission system is *congested*. Two types of locational pricing systems have been adopted in electricity markets for congestion management: zonal pricing and nodal pricing (or locational marginal pricing (LMP)).

words, a decision-maker's objective is to maximize the expected return and minimize the variance of the return. Combining the two, we can define the objective function or the utility function of a decision-maker in terms of the expected return $E(r)$ and variance of returns $\sigma^2(r)$ as follows [26]:

$$U = E(r) - \frac{1}{2}A\sigma^2(r) \quad (3)$$

where U is the utility function and A the weighting factor that reflects the decision maker's preference or aversion of risk. Positive A indicates a person is risk aversion, negative A indicates risk loving, and $A=0$ indicates the person is risk neutral. The larger is the value of A , the more risk aversion is the person. The determination of the exact value of the weighting factor A is typically the most difficult part of any theory trying to combine two objectives together. A broad range of studies, taking into account the full range of available assets, places the degree of risk aversion for the representative investor in the range of 2.0–4.0 [38]. In the financial text book [26], $A=3$ is used as average risk aversion, and consequently, $A>3$ for more risk aversion and $A<3$ for less risk aversion.

Using the above utility function, the optimal portfolio is obtained by maximizing the utility function with respect to the weights of assets:

$$\begin{aligned} \text{Max}_{w_i} U &= E(r_C) - \frac{1}{2}A\sigma^2(r_C) \\ \text{s.t.} \quad &\sum_{i=1}^{n+1} w_i = 1 \\ &w_i \geq 0 \end{aligned} \quad (4)$$

where $E(r_C)$ and $\sigma^2(r_C)$ are given by (1) and (2), respectively. The solution $\bar{w}_i (i=1-n+1)$ gives optimal allocation of assets into a risk-free asset $\bar{w}_{n+1}$ and n risky assets $\bar{w}_i (i=1-n)$. The optimal portfolio selection problem (4) is a quadratic programming problem. It, of course, can be solved directly using a standard quadratic programming solution algorithm [39] or a software package such as one available in Matlab. However, more insights can be gained and a graphical interpretation can be obtained if we solve it in two steps, following human intuition in solving this problem. Most individuals would divide the problem into two sub problems. They would first decide on what risky portfolio to invest, i.e., the allocation of risky investment into an optimal portfolio to diversify its risk. They would then consider the problem of how much money to take out from the bank to invest in risky investments, assuming money in the bank is considered risk-free.

In the Appendix A, we show mathematically that the solution of the portfolio optimization problem (4) is equivalent to the solutions of the following two optimization problems: one with n risky assets (5) and the other with one risk-free asset and one risky asset (6). The two-step approach is the standard approach followed in Portfolio Theory and is usually intuitively presented. We have not found a rigorous mathematical justification of such intuitive arguments as we proved in the Appendix A.

Step 1: Optimal risky portfolio

$$\begin{aligned} \text{Max}_{w_i} s &= \frac{E(r_P) - r_B}{\sigma(r_P)} \\ \text{s.t.} \quad &\sum_{i=1}^n w_i = 1 \\ &w_i \geq 0 \end{aligned} \quad (5)$$

where

$$\begin{aligned} E(r_P) &= \sum_{i=1}^n w_i E(r_i) \\ \sigma(r_P) &= \left(\sum_{i=1}^n w_i^2 \sigma_i^2 + \sum_{i \neq j} \sum w_i w_j \sigma_{ij} \right)^{1/2} \end{aligned}$$

The solution method to this optimization problem (5) can be found in [35]. Let the solution to the optimal risky portfolio problem be denoted by $w_i^* (i=1-n)$.

Step 2: Optimal allocation between a risk-free and a risky investment

$$\text{Max}_y U(y) = E(r_C) - \frac{1}{2}A\sigma^2(r_C) \quad (6)$$

where y is the investment proportion allocated to the risky asset, and

$$\begin{aligned} E(r_C) &= (1-y)r_B + yE(r_P^*) \\ \sigma^2(r_C) &= y^2 \sigma^2(r_P^*) \end{aligned}$$

Setting the derivative of this expression to zero and solving for y yields the optimal position for risk-averse investors in the risky asset, y^* , as follows:

$$y^* = \frac{E(r_P^*) - r_B}{A\sigma^2(r_P^*)} \quad (7)$$

where

$$\begin{aligned} E(r_P^*) &= \sum_{i=1}^n w_i^* E(r_i) \\ \sigma^2(r_P^*) &= \sum_{i=1}^n \sum_{j=1}^n w_i^* w_j^* \sigma_{ij}. \end{aligned}$$

Combining the solutions to the sub problems (5) and (6), we obtain the solution to the optimal portfolio selection (4) as follows:

$$\begin{aligned} \text{Risk-free asset, } \bar{w}_{n+1} &= 1 - y^*; \\ \text{Risky assets, } \bar{w}_i &= y^* w_i^* \quad (i=1-n). \end{aligned}$$

3.4. Graphical interpretation

The two-step solution has an intuitive graphical interpretation. Let us consider the optimal risky portfolio problem (5). The feasible set on the mean-standard deviation ($E(r)-\sigma$) diagram

is the points corresponding to all combinations of $w_i (i = 1-n)$ that satisfy:

$$E(r_P) = \sum_{i=1}^n w_i E(r_i)$$

$$\sigma(r_P) = \left(\sum_{i=1}^n w_i^2 \sigma_i^2 + \sum_{i \neq j} w_i w_j \sigma_{ij} \right)^{1/2}$$

$$\sum_{i=1}^n w_i = 1$$

$$w_i \geq 0$$

where $E(r_i) (i = 1-n)$ and $\sigma_{ij} (i, j = 1-n)$ are given.

It can be shown that the feasible set is connected and convex to the left [36], as showed in Fig. 1. Given an expected return value, a lowest portfolio variance can be attained through optimally selecting the weights of risky-assets. In the mean-standard deviation graph, given an expected return value, we can always find a portfolio with the given expected return and the lowest variance. All the portfolio points with such an attribute form a *frontier*. This frontier is called the *minimum-variance frontier* (showed in Fig. 1). The bottom part of the minimum-variance frontier is inefficient since for any portfolio in the lower portion of the minimum-variance frontier, there is a portfolio with the same standard deviation and a greater expected return positioned directly above it. Hence all the portfolios, which lie on the minimum-variance frontier from the minimum-variance portfolio and upward, provide the best risk-return combinations and thus are candidates for the optimal portfolio. These portfolios form the so called *efficient frontier* (see Fig. 1).

Assume that there is a risk-free asset denoted by B , portfolio selection can be achieved with two steps. The first step is to find the optimal risky portfolio. Any point in the efficient frontier represents a risky portfolio. A straight line connecting risk-free asset B and any risky portfolio in the efficient frontier is called Capital Allocation Line (CAL) (see Fig. 2). The slope of the CAL is the reward-to-risk ratio, a higher reward-to-risk ratio is preferred by the investor. Graphically, the optimal risky portfolio P , is the point where CAL is tangent to the efficient frontier since this CAL has the highest reward-to-risk ratio, i.e., the steepest

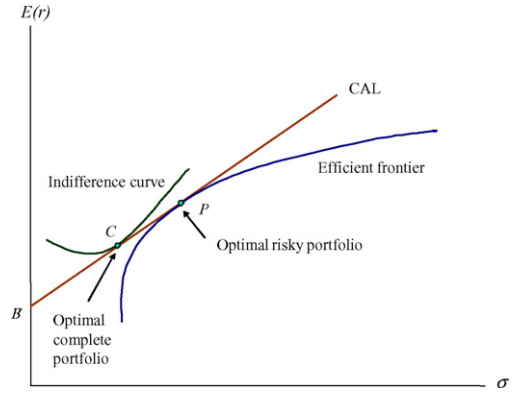


Fig. 2. Portfolio selection with a risk-free asset.

slope. This is precisely the Optimal Risky Portfolio Problem (5).

The second step of the portfolio selection is to allocate budget between the risk-free asset B and the optimal risky portfolio P . Graphically, the optimal combination is the point where CAL touches the highest value of U , i.e., it is tangent to the constant U curve, or the indifference curve, as showed in Fig. 2.

The presence of a risk-free asset makes it possible to solve the overall optimization problems (4) in two-steps (5) and (6). A remarkable implication of the two-step solution is that there is a separation principle. Note that the optimal risky portfolio selection problem (5) is independent of the decision-maker's risk preference A . In other words, the optimal risky portfolio on the efficient frontier can be determined once there is a reference point of a risk-free asset and this optimal portfolio is the same for all decision makers regardless of their risk preferences. Risk preference only comes in at the second step to decide on the proportion of the risk-free asset and this optimal risky portfolio.

In the absence of a risk-free asset, the determination of which point on the efficient frontier is optimal has to rely on a utility function such as in (4) that incorporates the decision-maker's risk preference. The optimization problem (4) can indeed be applied to solve the optimal portfolio selection without a risk-free asset.

4. Energy allocation between spot market and contract market

We now apply the portfolio optimization method presented in Section 3 to the energy allocation between spot market and contract market when the market adopts a LMP pricing scheme. Assume that a Genco trades energy through both spot market and bilateral contract market. For bilateral contracts, if there is congestion in the transmission system, the Genco will have to pay all or part of the congestion charges which depends on the specific market rules or on the negotiation between the Genco and its consumers. The congestion charge between any two locations is the product of the spot price difference between these two locations and the transmitted energy (quantity in MWh). Locational spot prices are uncertain and fluctuate. In such a case, only local contracts signed with local customers are risk-free trades, non-local contracts signed with non-local customers are risky trades due to the uncertainty in congestion charges. Therefore,

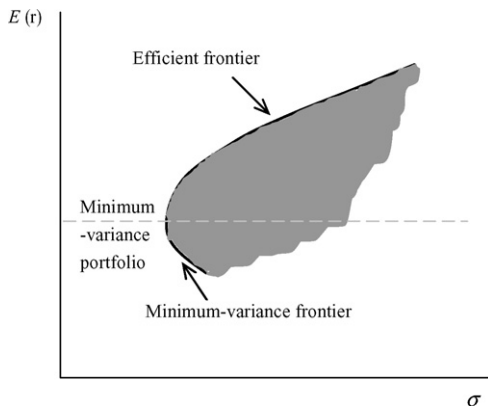


Fig. 1. Efficient frontier.

there are three types of trading approaches for a Genco: risk-free (local) contracts, risky (non-local) contracts and risky spot market.

Assume that there are n areas or pricing nodes in an electricity market. A Genco is located in Area 1; other areas are labeled from Area 2 to Area n . To simplify, suppose that the Genco could sign one bilateral contract with each area's customers at fixed energy price. Hence, the Genco has $n + 1$ potential transactions during the planning period,⁵ i.e., one risk-free local-bilateral contract, $n - 1$ risky non-local bilateral contracts and one risky transaction traded in the spot market. The question is how to allocate energy among these potential transactions in order to maximize profits with relatively low risk. Energy allocation without and with local bilateral contract will be considered respectively in the following.

4.1. Energy allocation without local bilateral contract

Without local bilateral contract, i.e., the risk-free option, a trading portfolio consists of only risky options and is therefore called a risky portfolio. The optimal risky portfolio can be achieved by solving optimization problem (4).

4.2. Energy allocation with local bilateral contract

If local bilateral contract (i.e., risk-free transaction) is available, the optimal trading portfolio can be obtained by solving optimization problem (4) directly or by solving problem (5) and (6) step-by-step.

Let us now specify the return characteristics of all trades (i.e., expected returns ($E(r_i)$), variances (σ_i^2) and covariances (σ_{ij})). Assume that the Genco has a quadratic cost-curve, $c(p, t, \lambda^F) = (a + bp + cp^2)t\lambda^F$, where p is the output power (MW); t the trading time of each trading interval (h); λ^F the fuel price (\$/MBtu) which is assumed given during the contract period; a , b , c are fuel consumption coefficients. There are M trading intervals in the trading period (i.e., contract period). The following notation will be used:

Cov: covariance,

E : expectation,

i : the index of the trading area or pricing node,

k : the index of the trading interval,

r_B : return on the local contract,

r_i : return on the i th trade, $i = 2-N$ denotes non-local bilateral contract signed with the i th Area's customers; $i = 1$ denotes the transaction traded in the spot market. Expectation and variance of r_i are denoted by $E(r_i)$ and σ_i^2 , respectively.

$\lambda_{i,k}^B$: the k th trading interval's contract price signed with customers of Area i .

λ_k^F : the k th trading interval's fuel price,

$\lambda_{i,k}^S$: the k th trading interval's spot price of Area i , the corresponding expectation is denoted by $E(\lambda_{i,k}^S)$.

4.2.1. Return characteristics of local contract

According to the definition of return, return = (revenue – cost)/cost, if all the energy are traded through the local contract,⁶ the corresponding return during the contract period is:

$$r_B = \frac{\sum_{k=1}^M p_k t \lambda_{1,k}^B - \sum_{k=1}^M (a + bp_k + cp_k^2) t \lambda_k^F}{\sum_{k=1}^M (a + bp_k + cp_k^2) t \lambda_k^F} = K \sum_{k=1}^M p_k t \lambda_{1,k}^B - 1 \quad (8)$$

where $K = 1/\sum_{k=1}^M (a + bp_k + cp_k^2) t \lambda_k^F$. $\lambda_{1,k}^B$ and λ_k^F are assumed certain when the Genco makes trading decisions, then $E(r_B) = r_B$, $\sigma^2(r_B) = 0$, i.e., local bilateral contract is a risk-free transaction.

Similarly, return characteristics of non-local contracts and spot markets can be expressed as follows.

4.2.2. Return characteristics of non-local contracts

The congestion charge is accounted when calculating the actual revenue on the non-local bilateral transaction. Generally, congestion charges should be paid by the associated bilateral transaction. But who (Gencos or energy purchasers) should pay how many percentage of the involved congestion charges depends on the specific market rules. That is, from a Genco's point of view, its congestion charge is between zero and the complete congestion charge of the associated bilateral transaction. In this paper, a factor β ($0 \leq \beta \leq 1$), which is decided by a specific electricity market, is used to denote the payment proportion of the Genco. Then we have:

$$E(r_i) = K \sum_{k=1}^M p_k t \left[\lambda_{i,k}^B - \beta(E(\lambda_{i,k}^S) - E(\lambda_{1,k}^S)) \right] - 1 \quad (i = 2-N) \quad (9)$$

$$\sigma_i^2 = K^2 \sum_{k=1}^M (p_k t \beta)^2 \left[\sigma^2(\lambda_{1,k}^S) + \sigma^2(\lambda_{i,k}^S) - 2\text{Cov}(\lambda_{1,k}^S, \lambda_{i,k}^S) \right] \quad (i = 2-N) \quad (10)$$

4.2.3. Return characteristics of spot markets

$$E(r_1) = K \sum_{k=1}^M p_k t \cdot E(\lambda_{1,k}^S) - 1 \quad (11)$$

$$\sigma_1^2 = K^2 \sum_{k=1}^M (p_k t)^2 \sigma^2(\lambda_{1,k}^S) \quad (12)$$

⁵ The planning period could be one day, one week, one month, one year or several years, etc.

⁶ Scheduled trading quantity of the energy of each transaction is the product of the corresponding allocation ration w_i and total energy (pt).

4.2.4. Correlations between risky trades

$$\begin{aligned}\sigma_{ij} &= \text{Cov}(r_i, r_j) \\ &= K^2 \sum_{k=1}^M (p_k t \beta)^2 \left[\sigma^2(\lambda_{1,k}^S) - \text{Cov}(\lambda_{1,k}^S, \lambda_{i,k}^S) \right. \\ &\quad \left. - \text{Cov}(\lambda_{1,k}^S, \lambda_{j,k}^S) + \text{Cov}(\lambda_{i,k}^S, \lambda_{j,k}^S) \right] \\ &\quad (i, j = 2-N)\end{aligned}\quad (13)$$

$$\begin{aligned}\sigma_{1i} &= \text{Cov}(r_1, r_i) = K^2 \sum_{k=1}^M (p_k t)^2 \beta \left[\sigma^2(\lambda_{1,k}^S) - \text{Cov}(\lambda_{1,k}^S, \lambda_{i,k}^S) \right] \\ &\quad (i = 2-N)\end{aligned}\quad (14)$$

$E(\lambda_{i,k}^S)$, $\sigma^2(\lambda_{i,k}^S)$ and $\text{Cov}(\lambda_{i,k}^S, \lambda_{j,k}^S)$ should be estimated before applying above equations. The estimation of these data is an applied area of research and is not the focus of this paper. In this paper, they are simply estimated based on historical data according to the statistical method.

Strictly speaking, when congestion occurs, the spot prices at various locations at that particular moment are related (through physical laws, i.e., Kirchoff laws and Ohms law, in the OPF). Over a period of time, however, such dependency might be captured statistically by the correlation coefficients or covariances of spot prices.

Example. We use PJM market data to illustrate the energy allocation methodology introduced in the last Section. In this example, three areas (pricing nodes) in an electricity market are considered. A Genco is located in Area 1. The Genco has four trading choices, i.e., one trading in the spot market, one bilateral contract signed with local customers (denoted by Contract 1), one bilateral contract signed with customers of Area 2 (denoted by Contract 2) and one bilateral contract signed with customers of Area 3 (denoted by Contract 3).

The trading period is one month and the trading time of each trading interval is one day or 24 h. The Genco's unit characteristics are: $a = 313.9102$ MBtu/h, $b = 7.6126$ MBtu/MWh, $c = 0.00199$ MBtu/MW²h, $p = 400$ MW. The historical data of daily spot price of Area 1, Area 2 and Area 3 are calculated based on the hourly price of month August from 1998 to 2005 in PENELEC, PEPCO and PECO, respectively (these data are available on the website of PJM [29]). Based on historical data, $E(\lambda_{i,k}^S)$, $\sigma^2(\lambda_{i,k}^S)$ and $\text{Cov}(\lambda_{i,k}^S, \lambda_{j,k}^S)$ ($i = 1-3$, $k = 1-31$) can be calculated with statistical method. Daily spot prices statistics of these three areas are thus obtained:

$$\begin{aligned}E(\lambda_1) &= 40.54 \text{ \$}/\text{MWh}, & \sigma(\lambda_1) &= 26.96 \text{ \$}/\text{MWh} (66.51\%), \\ \sigma(\lambda_2) &= 33.42 \text{ \$}/\text{MWh} (72.40\%), & E(\lambda_3) &= 46.44 \text{ \$}/\text{MWh},\end{aligned}$$

Let $p_i = 400$ MW, $\lambda_i^F = 3.0$ \$/MBtu, $\beta = 1$. Two cases are considered to demonstrate the impact of the risk aversion of the Genco and contract prices on the energy allocation or trading schedule.

Table 1

Optimal energy allocation ratios for Case 1

Risk aversion (A)	Energy allocation ratios				Utility value
	Contract 1	Contract 2	Contract 3	Spot market	
2	0.1460	0.2148	0.1245	0.5147	0.4336
3	0.4306	0.1432	0.0830	0.3432	0.4271
4	0.5730	0.1074	0.0622	0.2574	0.4239

Table 2

Optimal energy allocation ratios for Case 2

Risk aversion (A)	Energy allocation ratios				Utility value
	Contract 1	Contract 2	Contract 3	Spot market	
2	0.0271	0.3598	0.1864	0.4267	0.4446
3	0.3513	0.2399	0.1243	0.2845	0.4405
4	0.5136	0.1799	0.0932	0.2133	0.4384

Case 1. Suppose that $\lambda_1^B = 39$, $\lambda_2^B = 44.5$, $\lambda_3^B = 44.5$. With the proposed energy allocation approach, energy allocation ratios for different risk-aversion level are calculated and showed in Table 1. Simulation results indicate that energy allocation depends on the Genco's risk-aversion. More energy is allocated to more risky trade if the Genco is less risk-averse and vice versa, which is consistent with institution.

Case 2. Suppose that $\lambda_1^B = 39.5$, $\lambda_2^B = 45.2$, $\lambda_3^B = 45.3$. Optimal energy allocation ratios for different risk-aversion level are showed in Table 2. In this case, risk-aversion level has the same impact on energy allocation as in Case 1. But for the same risk-aversion level, more energy is allocated to the bilateral contracts compared to Case 1. The reason is that contract prices are higher in Case 2 than in Case 1. That is, for the same market situation, same decision-maker would trade more energy through bilateral contract market if contract prices are higher. It is also consistent with intuition.

5. Conclusion

We have formulated the general portfolio optimization problem as a quadratic programming (QP) problem. The problem can be solved numerically by a standard QP algorithm. Nevertheless, we link this problem to the standard approach in the financial literature. We show that the solution to the overall portfolio optimization problem with one risk-free and n risky assets can be obtained in two steps: first by optimal selection of n risky portfolio and then by optimal allocation between the risk-free asset and the risky portfolio obtained in the first step. We also include a brief tutorial treatment of the standard approach in

$$\begin{aligned}E(\lambda_2) &= 46.15 \text{ \$}/\text{MWh}, \\ \sigma(\lambda_3) &= 39.69 \text{ \$}/\text{MWh} (85.46\%).\end{aligned}$$

financial theory.

The general portfolio optimization methodology with n risky assets can be applied to energy allocation between spot market and bilateral contracts in a market where locational pricing,

either zonal or nodal, is adopted to mitigate transmission congestion. A Genco, while trading with a non-local customer, may pay congestion charge, which is a function of the locational price difference. Therefore, all non-local contracts are risky as a result of congestion charge. An example using PJM market data is used to illustrate the methodology. The results are consistent with intuition. The method, indeed, can be used to quantify the intuitive approach of allocating energy between markets. More accurate estimation on the statistics of spot prices (i.e., $E(\lambda_{i,k}^S)$, $\sigma^2(\lambda_{i,k}^S)$ and $\text{Cov}(\lambda_{i,k}^S, \lambda_{j,k}^S)$) makes the trading schedule more applicable.

The methodology presented here is general and can be applied in a more sophisticated manner to more detailed practical problems than the example in the paper illustrates.

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Appendix A. Theorem of portfolio optimization and its proof

The utility function defined by $U(\cdot) = E(r) - (1/2)A \cdot \sigma^2(r)$, is convex. The optimal portfolio can be achieved by directly maximizing the utility function with respect to the weights y and $w_i (i = 1-n)$, i.e.,

$$\begin{aligned} \text{Max}_{y, w_i} U(\cdot) &= E(r_C) - \frac{1}{2} A \sigma^2(r_C) \\ \text{s.t.} \quad \sum_{i=1}^n w_i &= 1 \\ w_i &\geq 0 \quad (i = 1, \dots, n) \end{aligned} \quad (\text{A.1})$$

where

$$\begin{aligned} E(r_C) &= (1-y)r_B + yE(r_P) \\ \sigma^2(r_C) &= y^2 \sigma^2(r_P) \\ E(r_P) &= \sum_{i=1}^n w_i E(r_i) \\ \sigma^2(r_P) &= \sum_{i=1}^n \sum_{j=1}^n w_i w_j \sigma_{ij} \end{aligned}$$

We call problem (A.1) the *Overall Portfolio Optimization Problem*. We are going to show that the Overall Portfolio Optimization problem can be solved in two steps.

Step 1 (Optimal risky portfolio)

Find the optimal risky portfolio $w_i (i = 1-n)$ by maximizing the slope of the CAL with respect to variables $w_i (i = 1-n)$,

i.e.,

$$\begin{aligned} \text{Max}_{w_i} s &= \frac{E(r_P) - r_B}{\sigma(r_P)} \\ \text{s.t.} \quad \sum_{i=1}^n w_i &= 1 \\ w_i &\geq 0 \quad (i = 1, \dots, n) \end{aligned} \quad (\text{A.2})$$

where

$$\begin{aligned} E(r_P) &= \sum_{i=1}^n w_i E(r_i) \\ \sigma(r_P) &= \sqrt{\sigma^2(r_P)} = \sqrt{\sum_{i=1}^n \sum_{j=1}^n w_i w_j \sigma_{ij}} \end{aligned}$$

Denote the optimal solution of (A.2) by w_i^* , and the corresponding values of r_P , $E(r_P)$ and $\sigma^2(r_P)$ are $r_P = \sum_{i=1}^n w_i^* r_i$, $E(r_P^*) = \sum_{i=1}^n w_i^* E(r_i)$ and $\sigma^2(r_P^*) = \sum_{i=1}^n \sum_{j=1}^n w_i^* w_j^* \sigma_{ij}$, respectively.

Step-2 (Optimal allocation between risk-free and risky investments)

Find the optimal allocation y between a risk-free investment and a risky investment by maximizing the utility function with respect to the variable y , i.e.,

$$\text{Max}_y U(\cdot) = E(r_C) - \frac{1}{2} A \sigma^2(r_C) \quad (\text{A.3})$$

where

$$\begin{aligned} E(r_C) &= (1-y)r_B + yE(r_P^*) \\ \sigma^2(r_C) &= y^2 \sigma^2(r_P^*) \end{aligned}$$

Denote the optimal solution to (A.3) by y^* .

The following theorem shows that the solutions to problems (A.2) and (A.3), taken together, form the solution to problem (A.1).

Theorem A.1. Suppose that $w_i^* (i = 1, \dots, n)$ and y^* are solutions to problem (A.2) and problem (A.3), respectively, then (w_i^*, y^*) is a solution to problem (A.1).

Proof. The three optimization problems are all convex. We are going to show that the Kuhn-Tucker optimality conditions for problems (A.2) and (A.3) together are equivalent to the optimality conditions to problem (A.1). Thus, the optimal solutions of (A.2) and (A.3) are the optimal solution to (A.1). $\square$

Denote the Lagrangian function of (A.2) by

$$L_1(w_i, \lambda, \mu) = \frac{E(r_P) - r_B}{\sigma(r_P)} + \lambda \sum_{i=1}^n w_i - \mu^T w,$$

where $\lambda \in R$, $\mu \in R^n$.

Since w^* is a solution to (A.2), according to the optimal conditions, there exists multipliers λ^* , μ^* such that (w^*, λ^*, μ^*)

satisfies

$$\begin{aligned} \partial_w L(w^*, \lambda^*, \mu^*) \\ = \frac{\xi_1}{\sigma(r_p^*)} - \frac{E(r_p^*) - r_B}{\sigma(r_p^*)} \cdot \frac{\xi_2^*}{2\sigma^2(r_p^*)} + \lambda^* e - \mu^* = 0, \\ \sum_{i=1}^n w_i^* = 1, \quad w_i^* \geq 0, \quad \mu_i^* \geq 0, \quad w_i^* \mu_i^* = 0 \end{aligned} \quad (\text{A.4})$$

where $e = (1, 1, \dots, 1)^T$, $\xi_1 = (E(r_1), E(r_2), \dots, E(r_n))^T$ and

$$\xi_2 = \begin{pmatrix} 2\sigma_{11}w_1 + (\sigma_{12} + \sigma_{21})w_2 + \dots + (\sigma_{1n} + \sigma_{n1})w_n \\ (\sigma_{21} + \sigma_{12})w_1 + 2\sigma_{22}w_2 + \dots + (\sigma_{2n} + \sigma_{n2})w_n \\ \dots \\ (\sigma_{n1} + \sigma_{1n})w_1 + (\sigma_{n2} + \sigma_{2n})w_2 + \dots + 2\sigma_{nn}w_n \end{pmatrix}$$

The ξ_2^* means the value of ξ_2 at point w^* .

The solution to (A.3) is obtained directly by solving $\partial_y U(\cdot) = 0$ as follows:

$$y^* = \frac{E(r_p^*) - r_B}{A\sigma^2(r_p^*)} \quad (\text{A.5})$$

(A.4) and (A.5) can be written equivalently

$$\begin{aligned} \frac{E(r_p^*) - r_B}{A\sigma(r_p^*)} \frac{\xi_1}{\sigma(r_p^*)} - \frac{(E(r_p^*) - r_B)^2}{A\sigma^2(r_p^*)} \frac{\xi_2^*}{2\sigma^2(r_p^*)} + \frac{E(r_p^*) - r_B}{A\sigma(r_p^*)} \lambda^* e \\ - \frac{E(r_p^*) - r_B}{A\sigma(r_p^*)} \mu^* = 0, \quad \sum_{i=1}^n w_i^* = 1, \quad w_i^* \geq 0, \\ \frac{E(r_p^*) - r_B}{A\sigma(r_p^*)} \mu_i^* \geq 0, \quad w_i^* \frac{E(r_p^*) - r_B}{A\sigma(r_p^*)} \mu_i^* = 0, \\ y^* = \frac{E(r_p^*) - r_B}{A\sigma^2(r_p^*)} \end{aligned} \quad (\text{A.6})$$

On the other hand, denote the Lagrangian function of problem (A.1) by

$$L(w, y, \lambda, \mu) = E(r_C) - \frac{1}{2} A\sigma^2(r_C) + \lambda \sum_{i=1}^n w_i - \mu^T w$$

Then the first-order necessary optimal condition can be written as a point (w, y, λ, μ) which satisfies

$$\begin{aligned} \nabla_w L(w, y, \lambda, \mu) = y\xi_1 - \frac{1}{2} Ay^2 \xi_2 + \lambda e - \mu = 0, \\ \partial_y L(w, y, \lambda, \mu) = -r_B + E(r_p) - Ay\sigma^2(r_p) = 0, \\ \sum_{i=1}^n w_i = 1, \quad w_i \geq 0, \quad \mu_i \geq 0, \quad w_i \mu_i = 0, \quad (i = 1, \dots, n) \end{aligned} \quad (\text{A.7})$$

where e, ξ_1, ξ_2 are defined as before. From the second expression of (A.7), y is solved directly and is then substituted to the first expression of (A.7). Therefore (A.7) follows

$$\begin{aligned} \nabla_w L(w, y, \lambda, \mu) \\ = \frac{E(r_p) - r_B}{A\sigma(r_p)} \times \frac{\xi_1}{\sigma(r_p)} - \frac{(E(r_p) - r_B)^2}{A\sigma^2(r_p)} \frac{\xi_2^*}{2\sigma^2(r_p)} \\ + \lambda e - \mu = 0, \quad \sum_{i=1}^n w_i = 1, \quad w_i \geq 0, \quad \mu_i \geq 0, \\ w_i \mu_i = 0, \quad (i = 1, \dots, n) \quad y = \frac{E(r_p) - r_B}{A\sigma^2(r_p)} \end{aligned} \quad (\text{A.8})$$

Comparing with the expression of (A.6) and (A.8), it shows that

$$z^* \equiv \left(w^*, y^*, \frac{E(r_p^*) - r_B}{A\sigma(r_p^*)} \lambda^*, \frac{E(r_p^*) - r_B}{A\sigma(r_p^*)} \mu^* \right)$$

is a solution to (A.8).

From the convex property of problem (A.1), the corresponding variables (w^*, y^*) of z^* is a solution to (A.1).

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